

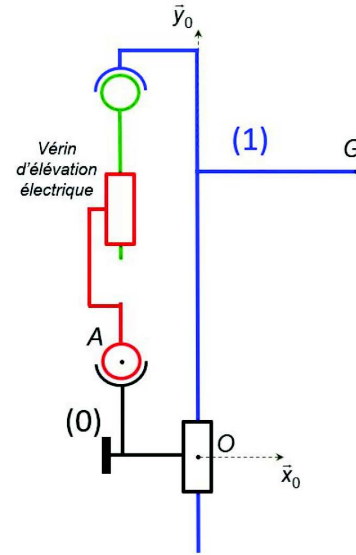
# INGÉSUP – TECHNICAL SYSTEMS 2

## *TD 4 – Mechanical actions*

### *(Solutions)*

ESME Bordeaux-Lille-Lyon-Paris

#### Exercise 1 : Maxi Move Patient Lifter [1]



#### Introduction

A patient lifter, employed in retirement homes and hospitals, is a mechanical help which allow the transfer of persons who cannot walk/move by themselves. The main use of the patient lifter is to move a patient from its bed to his/her chair and vice versa.

This mechanical system, which is illustrated by its kinematic diagram above, is composed of:

- a frame **0** associated to the reference frame  $R_0(O, \vec{x}_0, \vec{y}_0, \vec{z}_0)$
- an assembly **1** (arm + patient), associated to the reference frame  $R_1(O, \vec{x}_1, \vec{y}_1, \vec{z}_1)$ , of maximum mass  $m$  and with a center mass  $G$  such as  $\vec{OG} = a\vec{x}_0 + y\vec{y}_0$
- an electrical lifting cylinder which allow the vertical displacement of the assembly 1. We indicate  $F_{01}$  the force created by the lifting cylinder, and  $\vec{OA} = -b\vec{x}_0 + c\vec{y}_0$ .
- a compression spring located between **0** and **1**, which allow to compensate a part of the patient weight (it pushes towards  $+\vec{y}_0$  and alleviates the lifting cylinder) such as  $k$  is the stiffness,  $y$  its length,  $y_0$  its initial length and  $(O, \vec{y}_0)$  its line of action.

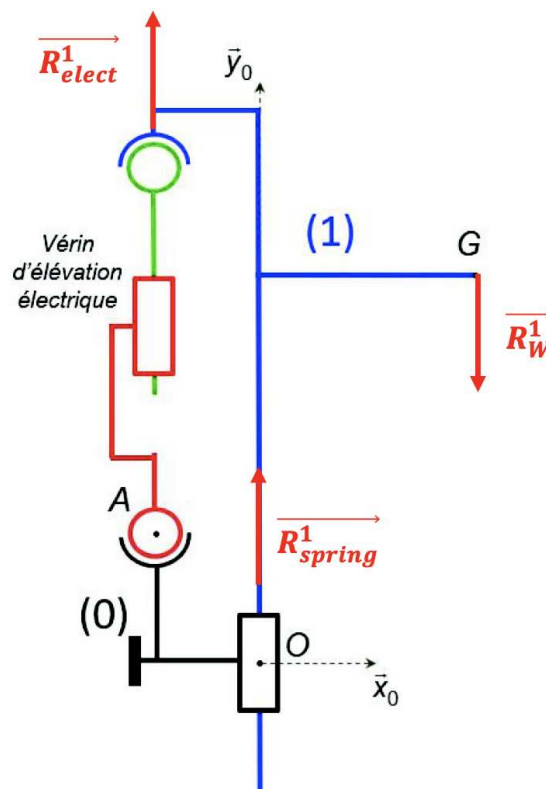
### Assumptions

All kinematic joints are without friction. The body weight and the lifting cylinder rod weight are neglected compared to the other mechanical actions.

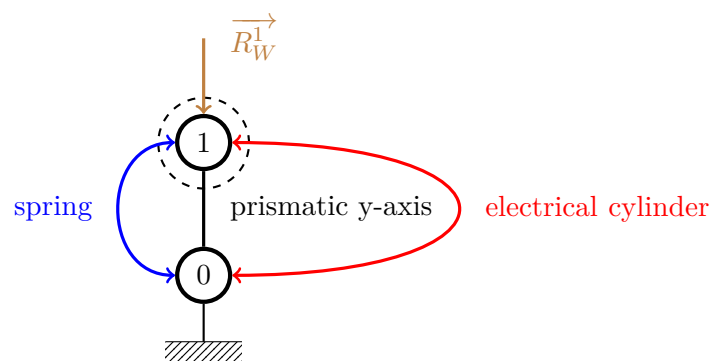
### Objectives

To be able to design correctly all components of the system, we need to determine the mechanical action from the kinematic joint between **1** and **0**.

**Q1.** Indicate on the kinematic diagram the mechanical action resultants of the lifter cylinder, of the spring and of the weight



**Q2.** Draw the mechanical action diagram.



**Q3. Isolate 1 and determine, when the mechanism is in equilibrium, the mechanical action torsor of the kinematic joint between 1 et 0.**

By isolating solid 1, the mechanical actions applied on it are:

- Weight  $\{\tau_W^1\}_G = \begin{Bmatrix} -mg\vec{Y}_0 \\ \vec{0} \end{Bmatrix}$
- Electrical device  $\{\tau_{elec}^1\}_A = \begin{Bmatrix} F_{01}\vec{Y}_0 \\ \vec{0} \end{Bmatrix}$
- Spring  $\{\tau_{spring}^1\}_O = \begin{Bmatrix} -k(y - y_0)\vec{Y}_0 \\ \vec{0} \end{Bmatrix}$
- Prismatic connection  $\{\tau_0^1\}_O = \begin{Bmatrix} X_{01}\vec{X}_0 + Z_{01}\vec{Z}_0 \\ L_{01}\vec{X}_0 + M_{01}\vec{Y}_0 + N_{01}\vec{Z}_0 \end{Bmatrix}$

The solid 1 is in equilibrium so we can apply the fundamental principle of statics:

$$\{\tau_{\Sigma}^{\Sigma}\} = \{0\}$$

$$\{\tau_W^1\}_O + \{\tau_{elec}^1\}_O + \{\tau_{spring}^1\}_O + \{\tau_0^1\}_O = \{0\}$$

*All torsors must be expressed at the same point and in the same coordinate system.*

Resultant equilibrium equation:

$$\vec{R}\{\tau_W^1\} + \vec{R}\{\tau_{elec}^1\} + \vec{R}\{\tau_{spring}^1\} + \vec{R}\{\tau_0^1\} = \vec{0}$$

$$\begin{bmatrix} 0 \\ -mg \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ F_{01} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -k(y - y_0) \\ 0 \end{bmatrix} + \begin{bmatrix} X_{01} \\ 0 \\ Z_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} X_{01} = 0 \\ F_{01} = -mg + k(y - y_0) \\ Z_{01} = 0 \end{cases}$$

Moment equilibrium equation:

$$\overrightarrow{MO}\{\tau_W^1\} + \overrightarrow{MO}\{\tau_{elec}^1\} + \overrightarrow{MO}\{\tau_{spring}^1\} + \overrightarrow{MO}\{\tau_0^1\} = \vec{0}$$

Where,

$$\begin{aligned} \overrightarrow{MO}\{\tau_W^1\} &= \overrightarrow{MG}\{\tau_W^1\} + \overrightarrow{OG} \times \vec{R}\{\tau_W^1\} \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} a \\ y \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ -mg \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ -amg \end{bmatrix} \end{aligned}$$

And,

$$\begin{aligned}
\overrightarrow{M_O}\{\tau_{elec}^1\} &= \overrightarrow{M_A}\{\tau_{elec}^1\} + \overrightarrow{OA} \times \overrightarrow{R}\{\tau_{elec}^1\} \\
&= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -b \\ c \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ F_{01} \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ 0 \\ -bF_{01} \end{bmatrix}
\end{aligned}$$

Therefore,

$$\begin{bmatrix} 0 \\ 0 \\ -amg \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -bF_{01} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} L_{01} \\ M_{01} \\ N_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} L_{01} &= 0 \\ M_{01} &= 0 \\ N_{01} &= amg + bF_{01} \end{cases}$$

In conclusion, the mechanical action torsor of the kinematic joint of 0 on 1 is,

$$\{\tau_0^1\}_O = \left\{ \begin{matrix} \vec{0} \\ (amg + bF_{01})\vec{z}_0 \end{matrix} \right\}$$

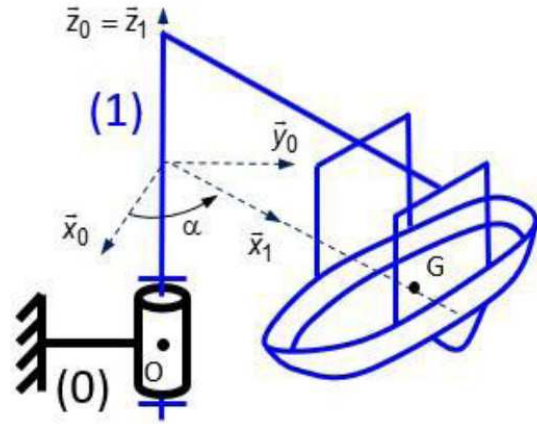
**Q4. Indicate which equation from the equilibrium equations is useful to determine the force required by the lifting cylinder to keep the system in equilibrium.**

The force required by the lifting cylinder to keep the system in equilibrium is:

$$F_{01} = mg + k(y - y_0)$$

As  $y$  is lower than  $y_0$ , therefore  $F_{01}$  is lower than  $mg$ .

## Exercise 2: Boat carrying console [1]



### Introduction

We are focusing on a boat carrying console which is used to get the boat into the water as well as to get it out of the water from a marina deck. This mechanical system, illustrated above, is composed of:

- a frame **0** associated to the reference frame  $R_0(O, \vec{x}_0, \vec{y}_0, \vec{z}_0)$ ;
- an assembly **1** (console + cables + boat), associated to the reference frame  $R_1(O, \vec{x}_1, \vec{y}_1, \vec{z}_1)$ , of maximum mass  $m = 4000 \text{ kg}$ , and with a center mass  $G$  located at

$$\vec{OG} = a\vec{x}_1 + b\vec{z}_1$$

with  $a = 6 \text{ m}$  and  $b = 4 \text{ m}$ .

The hinge joint between **0** and **1** is motorized. We indicate  $C_{01}$  the torque provided by the motor.

### Assumptions

All kinematic joints are without friction. A fluid dynamic study has been led to determine the mechanical action of the wind on the boat in the worst conditions, the torsor is:

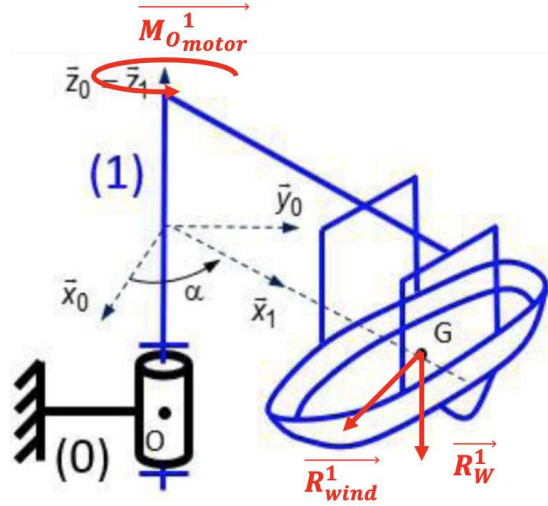
$$\{\tau_{wind}^1\} = \left\{ \begin{matrix} -F_{wind}\vec{y}_1 \\ \vec{0} \end{matrix} \right\}_G$$

where  $F_{wind} = 1500 \text{ daN}$

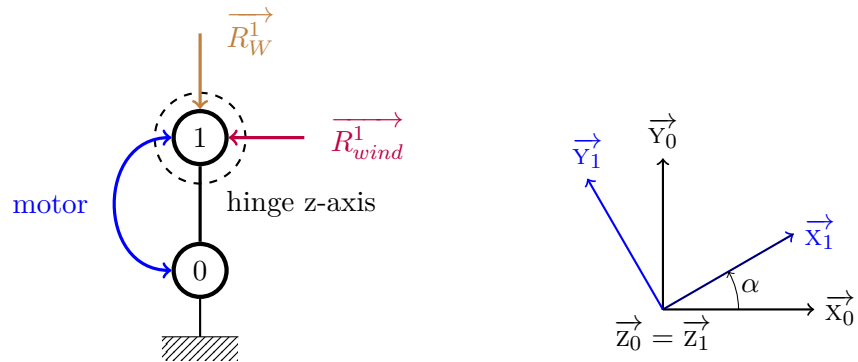
### Objectives

To be able to design correctly all components of the system, we need to determine the mechanical action from the kinematic joint between **1** and **0**.

Q1. Indicate on the kinematic diagram the mechanical action resultants of the wind, of the weight and of the torque provided by the motor



Q2. Draw the mechanical action diagram and the basis change figure.



Q3. Isolate 1 and determine, when the mechanism is in equilibrium, the mechanical action torsor of the kinematic joint between 1 et 0 expressed in basis 1. A numerical application is required.

By isolating solid 1, the mechanical actions applied on it are:

- Wind  $\{\tau_{wind}^1\}_G = \begin{Bmatrix} -F_{wind}\vec{Y}_1 \\ \vec{0} \end{Bmatrix}$
- Weight  $\{\tau_W^1\}_G = \begin{Bmatrix} -mg\vec{Z}_0 \\ \vec{0} \end{Bmatrix}$
- Motor  $\{\tau_{motor}^1\}_O = \begin{Bmatrix} \vec{0} \\ C_{01}\vec{Z}_0 \end{Bmatrix}$

- Hinge connection  $\{\tau_0^1\}_O = \begin{Bmatrix} X_{01}\vec{X}_1 + Y_{01}\vec{Y}_1 + Z_{01}\vec{Z}_1 \\ L_{01}\vec{X}_1 + M_{01}\vec{Y}_1 \end{Bmatrix}$

The solid 1 is in equilibrium so we can apply the fundamental principle of statics:

$$\{\tau_{\vec{\Sigma}}^{\Sigma}\} = \{0\}$$

$$\{\tau_{wind}^1\}_G + \{\tau_W^1\}_G + \{\tau_{motor}^1\}_O + \{\tau_0^1\}_O = \{0\}$$

*All torsors must be expressed at the same point and in the same coordinate system.*

Resultant equilibrium equation:

$$\vec{R}\{\tau_{wind}^1\} + \vec{R}\{\tau_W^1\} + \vec{R}\{\tau_{motor}^1\} + \vec{R}\{\tau_0^1\} = \vec{0}$$

$$\begin{bmatrix} 0 \\ -F_{wind} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} X_{01} \\ Y_{01} \\ Z_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} X_{01} = 0 \\ Y_{01} = F_{wind} \\ Z_{01} = mg \end{cases}$$

Moment equilibrium equation:

$$\vec{M}_O\{\tau_{wind}^1\} + \vec{M}_O\{\tau_W^1\} + \vec{M}_O\{\tau_{motor}^1\} + \vec{M}_O\{\tau_0^1\} = \vec{0}$$

Where,

$$\begin{aligned} \vec{M}_O\{\tau_{wind}^1\} &= \vec{M}_G\{\tau_{wind}^1\} + \vec{OG} \times \vec{R}\{\tau_{wind}^1\} \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} a \\ 0 \\ b \end{bmatrix} \times \begin{bmatrix} 0 \\ -F_{wind} \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} bF_{wind} \\ 0 \\ -aF_{wind} \end{bmatrix} \end{aligned}$$

And,

$$\begin{aligned} \vec{M}_O\{\tau_W^1\} &= \vec{M}_G\{\tau_W^1\} + \vec{OG} \times \vec{R}\{\tau_W^1\} \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} a \\ 0 \\ b \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ amg \\ 0 \end{bmatrix} \end{aligned}$$

Therefore,

$$\begin{bmatrix} bF_{wind} \\ 0 \\ -aF_{wind} \end{bmatrix} + \begin{bmatrix} 0 \\ amg \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ C_{01} \end{bmatrix} + \begin{bmatrix} L_{01} \\ M_{01} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} L_{01} = -bF_{wind} \\ M_{01} = -amg \\ C_{01} = aF_{wind} \end{cases}$$

In conclusion, the mechanical action torsor of the kinematic joint of 0 on 1 is,

$$\{\tau_0^1\}_O = \begin{Bmatrix} F_{wind}\vec{Y}_1 + mg\vec{Z}_1 \\ -bF_{wind}\vec{X}_1 - amg\vec{Y}_1 \end{Bmatrix}$$

The numerical application gives you:

$$\{\tau_0^1\}_O = \left\{ \begin{array}{l} 15000\vec{Y}_1 + 39240\vec{Z}_1 \\ -60000\vec{X}_1 - 235440\vec{Y}_1 \end{array} \right\}$$

Where the force is expressed in N and the moment in N m.

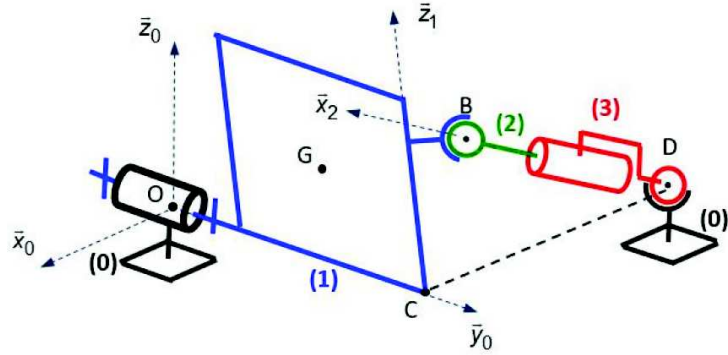
**Q4. Indicate which equation from the equilibrium equations is useful to determine the torque required by the motor to keep the system in equilibrium.**

The torque required from the motor to keep the system in equilibrium is:

$$C_{01} = aF_{wind}$$

It is given by the fundamental principle of statics for the moment. As  $C_{01}$  is positive, therefore the direction of rotation is in positive  $\vec{Z}_1$ . We can notice the torque is independent from the angle  $\alpha$ . It is not the case in every problem.

### Exercise 3 : Smoke extraction damper [1]



#### Introduction

Buildings with a surface higher than  $300 \text{ m}^2$  must have smoke extraction dampers, to extract smoke in case of a fire. Following the alarm, these dampers must be open automatically. They can be located on the ceiling or they can be windows. We will study a smoke extraction damper as a window. This mechanical system, illustrated above, is composed of:

- a frame **0** associated to the reference frame  $R_0(O, \vec{x}_0, \vec{y}_0, \vec{z}_0)$
- a damper **1**, associated to the reference frame  $R_1(O, \vec{x}_1, \vec{y}_1, \vec{z}_1)$ , such as  $\theta = (\vec{z}_0, \vec{z}_1)$ , of mass  $m$  and of center mass  $G$  such as  $\vec{OG} = a\vec{y}_1 + h\vec{z}_1$
- a jack **2-3** which allow the damper rotation **1**, associated to the reference frame  $R_2(D, \vec{x}_2, \vec{y}_2, \vec{z}_2)$  such as  $\alpha = (\vec{x}_0, \vec{x}_2)$ . We indicate  $F_{01}$  the force provided by the jack, and  $\vec{CD} = -e\vec{x}_0$  and  $\vec{OC} = a\vec{y}_0$
- a torsion spring located between **0** and **1**, allowing to close the damper properly when no alarm is detected (it positions the window towards  $-\vec{y}_0$ ), such as  $k$  is the stiffness,  $\theta$  its angle, and  $\theta_0$  its initial angle.

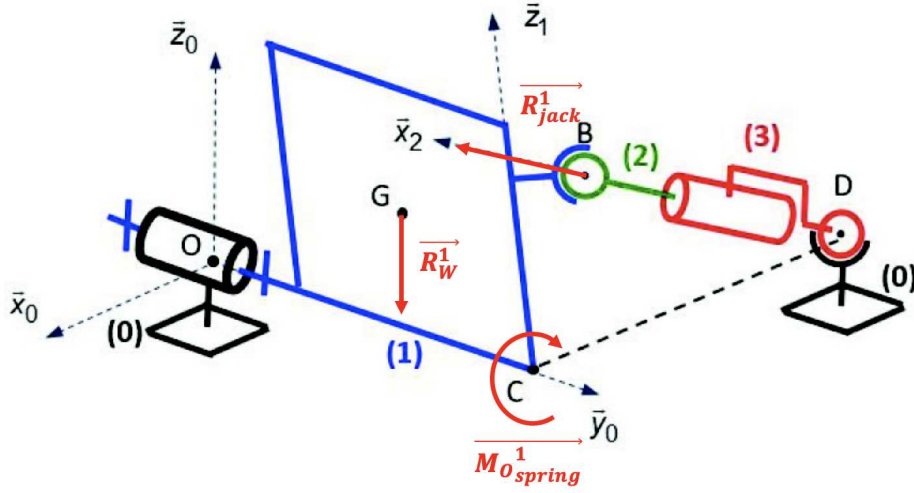
#### Assumptions

All kinematic joints are without friction. The body weight and the rod weight of the jack are neglected compared to the other mechanical actions.

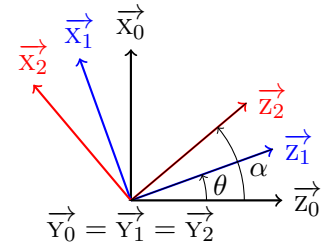
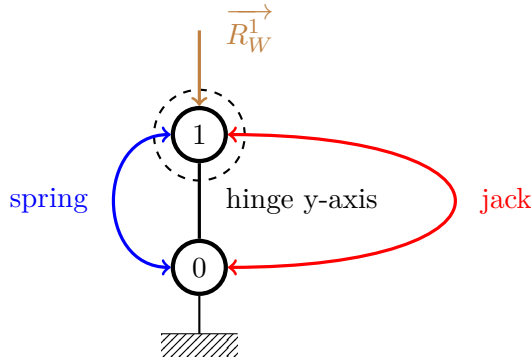
#### Objectives

To be able to design correctly all components of the system, we need to determine the mechanical action from the kinematic joint between **1** and **0**.

Q1. Indicate on the kinematic diagram the mechanical action resultants of the jack, of the weight and of the moment provided by the torsion spring.



Q2. Draw the mechanical action diagram and the basis change figures.



Q3. Isolate 1 and determine, when the mechanism is in equilibrium, the mechanical action torsor of the kinematic joint between 1 et 0 expressed in basis 1. Deduce its simpler expression.

By isolating solid 1, the mechanical actions applied on it are:

- Weight  $\{\tau_W^1\}_G = \begin{Bmatrix} -mg\vec{z}_0 \\ \vec{0} \end{Bmatrix}$
- Jack  $\{\tau_{jack}^1\}_D = \begin{Bmatrix} F_{01}\vec{x}_2 \\ \vec{0} \end{Bmatrix}$
- Torsion spring  $\{\tau_{spring}^1\}_O = \begin{Bmatrix} \vec{0} \\ -k(\theta - \theta_0)\vec{y}_0 \end{Bmatrix}$

- Hinge connection  $\{\tau_0^1\}_O = \begin{Bmatrix} X_{01}\vec{x}_0 + Y_{01}\vec{y}_0 + Z_{01}\vec{z}_0 \\ L_{01}\vec{x}_0 + N_{01}\vec{z}_0 \end{Bmatrix}$

The solid 1 is in equilibrium so we can apply the fundamental principle of statics:

$$\{\tau_{\Sigma}^{\Sigma}\} = \{0\}$$

$$\{\tau_W^1\}_G + \{\tau_{jack}^1\}_D + \{\tau_{spring}^1\}_O + \{\tau_0^1\}_O = \{0\}$$

All torsors must be expressed at the same point and in the same coordinate system.

Resultant equilibrium equation:

$$\vec{R}\{\tau_W^1\} + \vec{R}\{\tau_{jack}^1\} + \vec{R}\{\tau_{spring}^1\} + \vec{R}\{\tau_0^1\} = \vec{0}$$

$$\begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + \begin{bmatrix} F_{01}\cos(\alpha) \\ 0 \\ -F_{01}\sin(\alpha) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} X_{01} \\ Y_{01} \\ Z_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} X_{01} = -F_{01}\cos(\alpha) \\ Y_{01} = 0 \\ Z_{01} = mg + F_{01}\sin(\alpha) \end{cases}$$

Moment equilibrium equation:

$$\vec{M}_O\{\tau_W^1\} + \vec{M}_O\{\tau_{jack}^1\} + \vec{M}_O\{\tau_{spring}^1\} + \vec{M}_O\{\tau_0^1\} = \vec{0}$$

Where,

$$\begin{aligned} \vec{M}_O\{\tau_W^1\} &= \vec{M}_G\{\tau_W^1\} + \vec{OG} \times \vec{R}\{\tau_W^1\} \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} h\sin(\theta) \\ a \\ h\cos(\theta) \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} \\ &= \begin{bmatrix} -amg \\ mgh\sin(\theta) \\ 0 \end{bmatrix} \end{aligned}$$

And,

$$\begin{aligned} \vec{M}_O\{\tau_W^1\} &= \vec{M}_D\{\tau_{jack}^1\} + \vec{OD} \times \vec{R}\{\tau_{jack}^1\} \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -e \\ a \\ 0 \end{bmatrix} \times \begin{bmatrix} F_{01}\cos(\alpha) \\ 0 \\ -F_{01}\sin(\alpha) \end{bmatrix} \\ &= \begin{bmatrix} -aF_{01}\sin(\alpha) \\ -eF_{01}\sin(\alpha) \\ -aF_{01}\cos(\alpha) \end{bmatrix} \end{aligned}$$

Therefore,

$$\begin{bmatrix} -amg \\ mgh\sin(\theta) \\ 0 \end{bmatrix} + \begin{bmatrix} -aF_{01}\sin(\alpha) \\ -eF_{01}\sin(\alpha) \\ -aF_{01}\cos(\alpha) \end{bmatrix} + \begin{bmatrix} 0 \\ -k(\theta - \theta_0) \\ 0 \end{bmatrix} + \begin{bmatrix} L_{01} \\ 0 \\ N_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} L_{01} = amg + aF_{01}\sin(\alpha) \\ F_{01} = \frac{mgh\sin(\theta) - k(\theta - \theta_0)}{e\sin(\alpha)} \\ N_{01} = aF_{01}\cos(\alpha) \end{cases}$$

In conclusion, the mechanical action torsor of the kinematic joint of 0 on 1 is,

$$\{\tau_0^1\}_O = \left\{ \begin{array}{l} -F_{01}\cos(\alpha)\vec{x}_0 + [mg + F_{01}\sin(\alpha)]\vec{z}_0 \\ [amg + aF_{01}\sin(\alpha)]\vec{x}_0 + aF_{01}\cos(\alpha)\vec{z}_0 \end{array} \right\}$$

This torsor can also be written as,

$$\{\tau_0^1\}_O = \left\{ \begin{array}{l} mg\vec{z}_0 - F_{01}\vec{x}_2 \\ amg\vec{x}_0 + aF_{01}\vec{z}_2 \end{array} \right\}$$

**Q4. Indicate which equation from the equilibrium equations is useful to determine the force of the jack required to keep the system in equilibrium.**

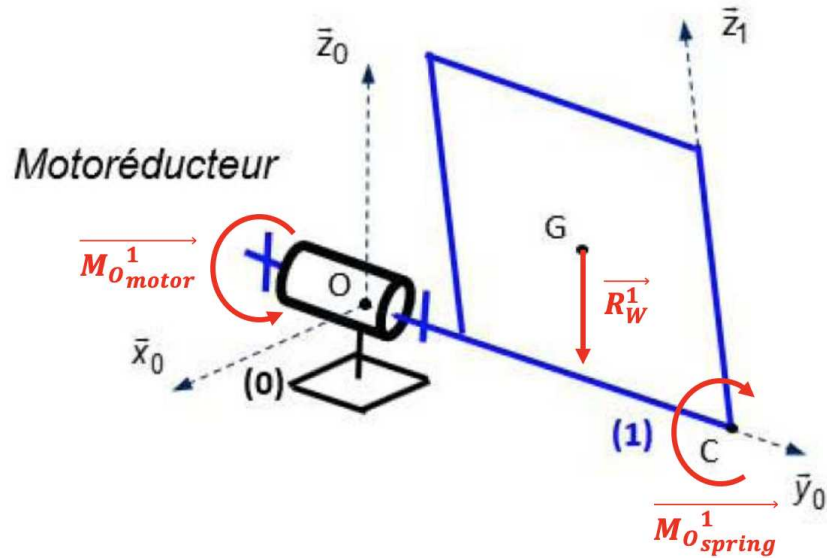
The force required from the jack to keep the system in equilibrium is:

$$F_{01} = \frac{mgh\sin(\theta) - k(\theta - \theta_0)}{e\sin(\alpha)}$$

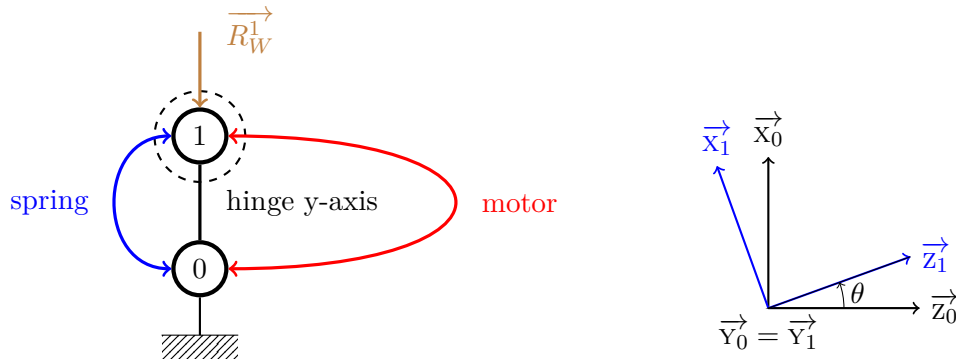
It is given by the fundamental principle of statics for the moment.

The system designer wish to replace the jack by a motor. We indicate  $C_{01}$  the torque provided by the motor.

**Q5. Indicate on the kinematic diagram the mechanical action resultants of the weight and of the moments provided by the torsion spring and the motor.**



**Q6. Draw the mechanical action diagram.**



**Q7. Isolate 1 and determine, when the mechanism is in equilibrium, the mechanical action torsor of the kinematic joint between 1 et 0.**

By isolating solid 1, the mechanical actions applied on it are:

- Weight  $\{\tau_W^1\}_G = \begin{Bmatrix} -mg\vec{z}_0 \\ \vec{0} \end{Bmatrix}$
- Motor  $\{\tau_{motor}^1\}_O = \begin{Bmatrix} \vec{0} \\ C_{01}\vec{y}_0 \end{Bmatrix}$
- Torsion spring  $\{\tau_{spring}^1\}_O = \begin{Bmatrix} \vec{0} \\ -k(\theta - \theta_0)\vec{y}_0 \end{Bmatrix}$
- Hinge connection  $\{\tau_0^1\}_O = \begin{Bmatrix} X_{01}\vec{x}_0 + Y_{01}\vec{y}_0 + Z_{01}\vec{z}_0 \\ L_{01}\vec{x}_0 + N_{01}\vec{z}_0 \end{Bmatrix}$

The solid 1 is in equilibrium so we can apply the fundamental principle of statics:

$$\{\tau_{\Sigma}^1\} = \{0\}$$

$$\{\tau_W^1\}_G + \{\tau_{motor}^1\}_O + \{\tau_{spring}^1\}_O + \{\tau_0^1\}_O = \{0\}$$

*All torsors must be expressed at the same point and in the same coordinate system.*

Resultant equilibrium equation:

$$\vec{R}\{\tau_W^1\} + \vec{R}\{\tau_{motor}^1\} + \vec{R}\{\tau_{spring}^1\} + \vec{R}\{\tau_0^1\} = \vec{0}$$

$$\begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} X_{01} \\ Y_{01} \\ Z_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} X_{01} = 0 \\ Y_{01} = 0 \\ Z_{01} = mg \end{cases}$$

Moment equilibrium equation:

$$\vec{M}_O\{\tau_W^1\} + \vec{M}_O\{\tau_{motor}^1\} + \vec{M}_O\{\tau_{spring}^1\} + \vec{M}_O\{\tau_0^1\} = \vec{0}$$

Where,

$$\begin{aligned}
\overrightarrow{M_O}\{\tau_W^1\} &= \overrightarrow{M_G}\{\tau_W^1\} + \overrightarrow{OG} \times \overrightarrow{R}\{\tau_W^1\} \\
&= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} h\sin(\theta) \\ a \\ h\cos(\theta) \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} \\
&= \begin{bmatrix} -amg \\ mgh\sin(\theta) \\ 0 \end{bmatrix}
\end{aligned}$$

Therefore,

$$\begin{bmatrix} -amg \\ mgh\sin(\theta) \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ C_{01} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -k(\theta - \theta_0) \\ 0 \end{bmatrix} + \begin{bmatrix} L_{01} \\ 0 \\ N_{01} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} L_{01} &= amg \\ C_{01} &= k(\theta - \theta_0) - mgh\sin(\theta) \\ N_{01} &= 0 \end{cases}$$

In conclusion, the mechanical action torsor of the kinematic joint of 0 on 1 is,

$$\{\tau_0^1\}_O = \left\{ \begin{matrix} mg\overrightarrow{Z_0} \\ amg\overrightarrow{X_0} \end{matrix} \right\}$$

**Q8. Indicate which equation from the equilibrium equations is useful to determine the torque required by the motor to keep the system in equilibrium.**

The torque required from the motor to keep the system in equilibrium is:

$$C_{01} = k(\theta - \theta_0) - mgh\sin(\theta)$$

It is given by the fundamental principle of statics for the moment.

## References

[1] <http://stephane.genouel.free.fr/>).